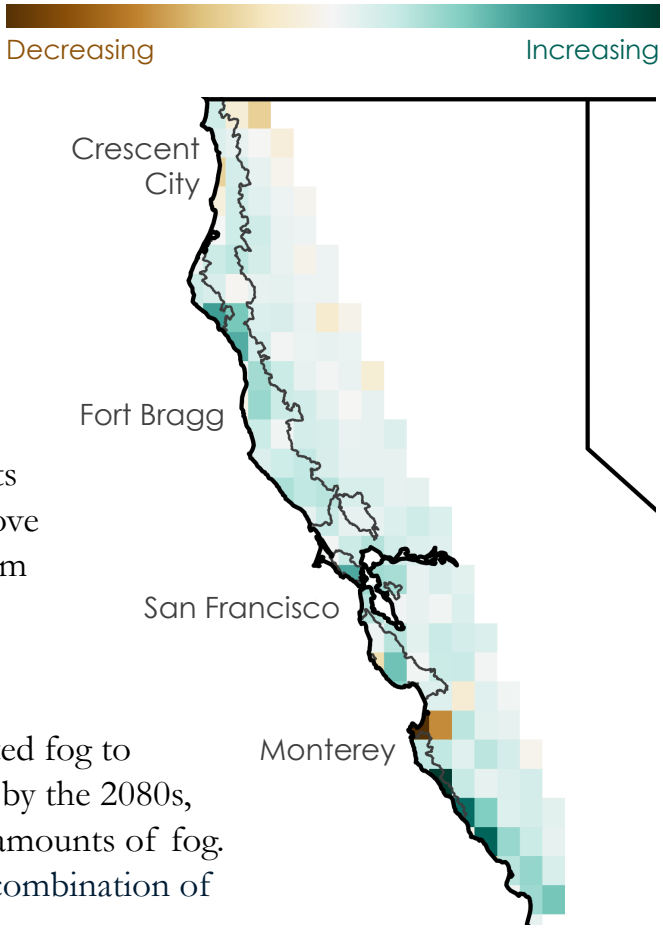
The Future of Fog: What's in Store for Coast Redwoods?

Redwoods Need Fog

During California summers, the north Pacific high pressure system pushes warm, moist air across the cooler California current, condensing the air and producing fog that blankets the coast for hundreds of miles. Redwoods take up to 30% of their moisture from this fog, so fog influences where redwood trees can live.

Projected Change in Average Hours of Fog per Day by 2090 (summertime)



The Future of Fog is Uncertain

Because redwood trees need fog, it's important to understand the impact of climate change on its formation. Previous studies have found mixed results about fog trends along the redwood range. To improve our understanding, a team of NASA researchers from the DEVELOP Program took a closer look.

New Insights: Good News for Redwoods

Using a machine learning model, researchers projected fog to *increase* across nearly the entire coast redwood range by the 2080s, particularly in places which currently have the least amounts of fog. Coastal fog dynamics are complex, influenced by a combination of large-scale climate patterns and local factors.

A Look into the Future

These results are the first future projections of fog, offering insights into what the California coast might look like 60 years from now. While climate change might increase fog prevalence, it can also increase other factors that stress redwood health, like fires and drought. Understanding future fog can make it easier for conservation organizations, like Save the Redwoods League, to manage these factors in tandem, planting trees in climate resilient areas.

This material is based upon work supported by NASA through contract 80LARC23FA024.

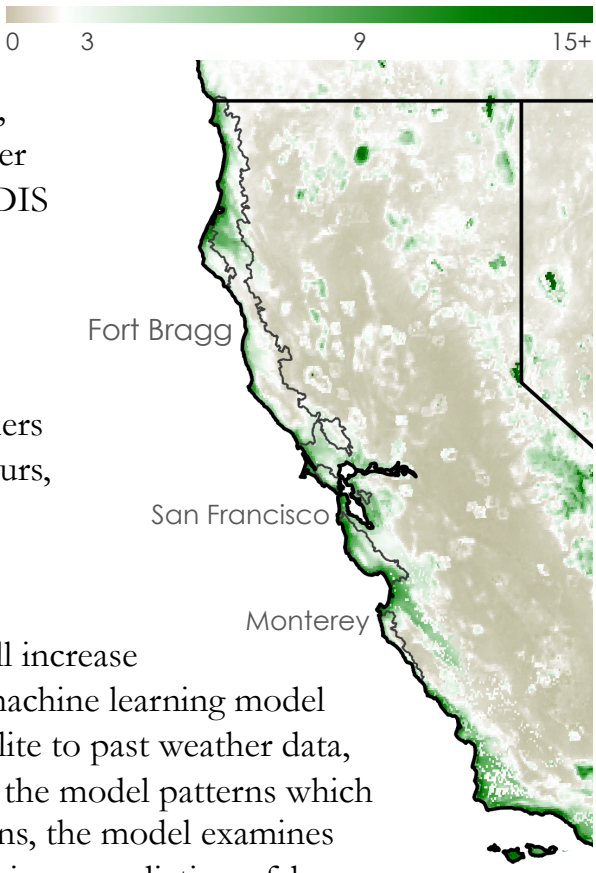
Into the Science

A Closer Look at How Researchers Identify and Predict Fog

Understanding Current Fog

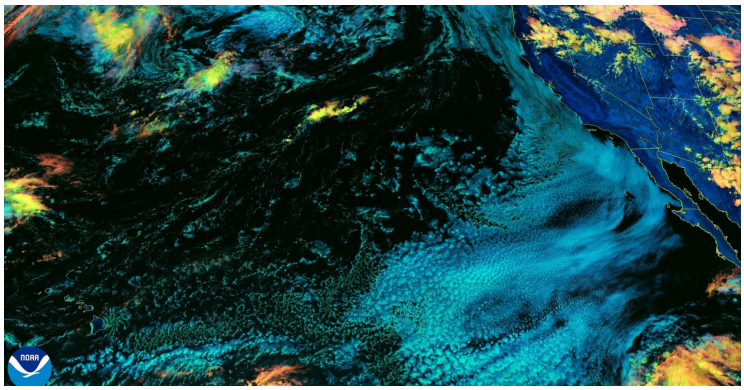
To predict future fog, it's important to understand fog's current presence along the California coast. To do this, the researchers used the NASA MODIS satellite, the GOES-18 weather satellite, and ground-based weather stations to map current fog conditions. The NASA MODIS land and sea skin temperature products provide useful context into the environmental conditions affecting fog formation. The GOES-18 satellite can tell if a cloud is made of liquid water (fog) or ice, which can be checked against the weather stations on the ground. The researchers added up fog presence over time, showing how many hours, on average, fog occurs at different locations.

Average Fog Hours per Day (2023–24)



How Do You Predict Fog?

Researchers used machine learning to project that fog will increase across much of the redwood range. To predict fog, the machine learning model compares where fog was detected using the GOES satellite to past weather data, like temperature, humidity, and wind speed. This teaches the model patterns which connect weather data to fog presence. Using these patterns, the model examines projected weather data from a NASA climate model, making a prediction of how much fog will occur in these future conditions.



An example scan from the GOES-18 satellite showing clouds consisting of liquid water (light blue) and ice crystals (yellow/orange).

Image Credit: NOAA (GOES-18 Cloud Top Phase)

Behind the Increase

As climate change warms our planet, more water from oceans and lakes will evaporate into the atmosphere, driving an increase in fog for the California coast. While the planet warms, this increase in fog would likely lead to cooler days along the coast, as the fog "insulates" the Earth's surface from the warm sunlight.

Overall, this research offers an exciting first look into the future of fog along the California coast and the redwood ecosystem.

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